

ECE-786 Programming Assignment - 2

Ashwin Mathew Olakangal

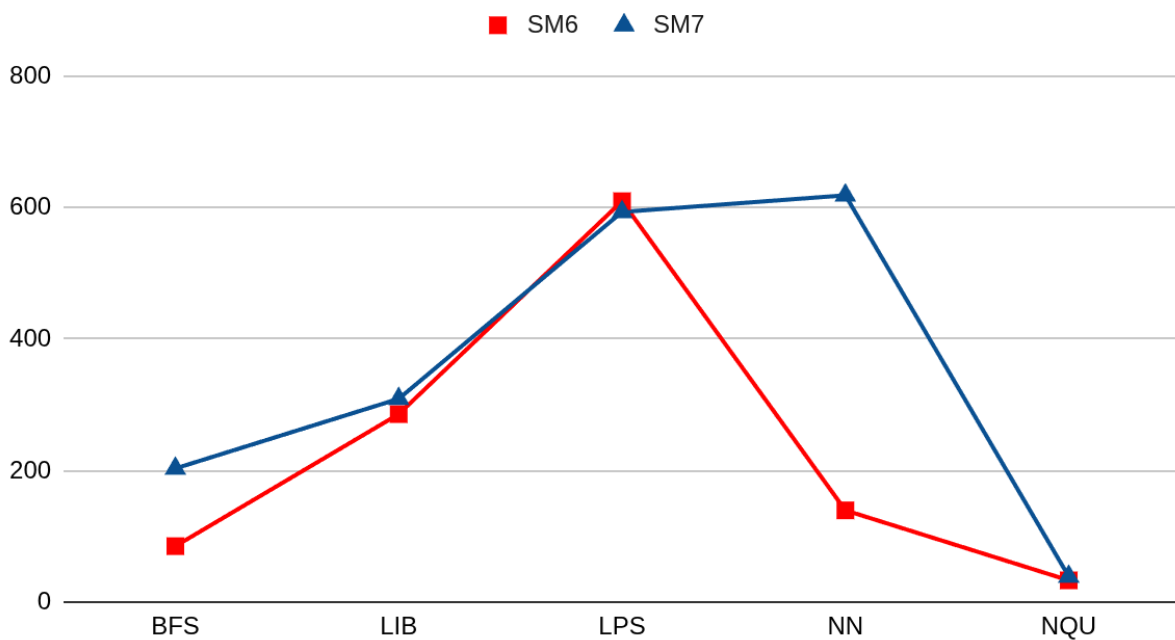
Task-1:

There was a single line that needed to be modified inside [instructions.cc](#) file. In particular we were interested in changing `add_impl()` function to execute `pow()` operation instead of F.P addition for 32-bit integers.

Task-2:

There wasn't a need to modify the files as the only goal was to run various benchmarks. The benchmarks were run across two configurations, the plots for which are attached below. We saw higher IPC across all the benchmarks in SM7 config compared to SM6 config.

benchmark (X-axis) V/s Avg. IPC (Y-axis)



Task-3:

We had to make significant changes across the following four files:

//update for task3 - comment wherever changes are made in the following files

shader.h -

We create new counters and initialize them to zero here in the header file (m_n_conditional_divergent_warps and m_n_conditional_branch_warps).

shader.cc -

Inside issue_warp() for total warps that see conditional branch we check if the next_inst->op is BRANCH_OP or not, since this is just a category of instructions we still need to distinguish between conditional branch and other types of branch such as uniform or indirect. To do this we check if the reconvp_pc is equal to -1 or not (branch conditional).

Inside issue_warp() for warps that have divergence in their branches we modify the updateSIMTStack() to return a boolean value (warp_divergence) to see if there is divergence in warp. Using this bool we check if the opcode is BRANCH_OP and warp_divergence == true. If both conditions are met we increment the counter.

Inside shader_core_stats::print() - updated to print the counter stats at the end of run.

abstract_hardware.h -

Change simt_stack::update() return type to bool

Change updateSIMTStack() return type to bool

abstract_hardware.cc -

Modify simt_stack::update() to return warp_divergence (which is a bool).

Modify updateSIMTStack() to return bool based on warp_divergence.

FOR LPS

total_conditional_divergent_warps = 54356

total_conditional_branch_warps = 411300